Swarm Witness: What Do Agent-Safety Mechanisms Actually Establish?

Tracks: 3 (Strengthening Agent Infrastructure), 4 (Multi-Agent Oversight and Control)

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Abstract

When agent-safety controls produce apparently valid evidence, what additional observations, assumptions, and comparisons are required before we can justify claims about their behavior? We introduce Swarm Witness, an executable claim-audit method that makes those distinctions explicit through source-pinned experiments, conventional reference monitors, and fail-closed findings. Classical authorization is the null hypothesis—not a new primitive we invent. We reproduce two established safety findings (R1 restart-safe authorization; R2 receipt-attribution ambiguity), compare a bounded BMO ledger property against a classical reference monitor (R3-L1), and test a three-way evidence classification on synthetic authorization and multi-principal scenarios (R4/R5). R4 separates missing evidence, failed enforcement, and supported conclusions, including a negative control that known-false grant observations cannot yield unqualified support. R5 transfers that taxonomy to a sequential three-principal shared-grant model (contention, A→B then B→C assignee hops, ledger-bound receipt misuse)—not concurrent live swarm execution. The contribution is reusable audit discipline, not a swarm authorization architecture or general BMO equivalence.

1. Introduction

Research question. When agent-safety controls produce apparently valid evidence, what additional observations, assumptions, and comparisons are required before we can justify claims about their behavior?

Authorization gates, receipt graphs, and retained grants are agent-facing controls that auditors must score carefully. A mechanism can work as implemented while the evidence used to justify broader claims remains insufficient. Swarm Witness makes that distinction testable through pinned reproductions, matched classical controls, and explicit observation boundaries. Classical authorization is the **null hypothesis**—not a new architecture we propose, and not a claim that BMO is equivalent to classical security.

Threat / failure modes. (i) An admission gate forgets an acknowledged rejection across restart when the durable log is not recovered before admit. (ii) A verifier treats possession of a copied receipt token as event attribution without exclusivity or authenticity. (iii) Retained start/grant metadata or an old receipt is treated as standing authority without consulting current grant state.

Main contributions (claimed).

1. **Pinned claim auditing** of two prior projects (R1/R2) as controlled exemplars under source pins—not as weekend discoveries of those mechanisms.
2. **New classical reference-monitor comparison** against BMO's transactional first-edit owner (r3-ledger-claim-v1), with a bounded L1 match under fail-closed native

qualification; legacy decision-model R3 stays unresolved.

3. **Three-way conclusion method (R4)** that distinguishes missing evidence, failed enforcement, and supported conclusions—including bounded principal→delegate handoff and a misbound-observation negative control (known-false evidence cannot yield unqualified support).
4. **Multi-principal method transfer (R5):** the same taxonomy on a synthetic sequential shared-grant model with named principals A/B/C (sequential contention, A→B then B→C assignee hops, ledger-bound receipt misuse). This is **not** concurrent worker execution or live BMO-mesh safety.

Tracks: **3** (Strengthening Agent Infrastructure) and **4** (Multi-Agent Oversight and Control). Preexisting BMO code and Oct 25 scaffold are marked in `PREEXISTING.md`; judged weekend work is the audit method, pins, baselines, experiments, and evidence packaging.

2. Related Work

We build on Apart Swarm Alignment themes of identity, durable commitments, and oversight under copy/rename/relay (Hammond et al., 2025; Chan et al., 2025; Greenblatt et al., 2024). Concrete targets:

- **SoundGate / "A Restart Is Not Permission"** — durable auth decisions and recovery-barrier repair (pinned package `soundgate==0.1.3`, source revision `6568403`). Original authors already reported the core repair; we reproduce under pin.
- **EvidenceGraph / Junejo "Receipt Possession Is Not Event Attribution"** — stress harness at git rev `a5ea5c9`; paper baseline rev `efbd419`. Again a bounded reproduction of a published information-quality failure.
- **BMO ordinary-retry retained grant** — preexisting durable grant / owner-witness surfaces under pinned `git_rev`; our new work is the matched ClassicalLedgerGate comparison and evidence packaging, not BMO itself.

In-repo baseline cards state inherited mechanism classes (recover-before-admit; no-implicit-exclusivity; durable grant-table checks) without equating every proxy to a decision match. Swarm Witness differs from building a new gate: it asks what labels survive matched baselines and discriminating checks.

3. Methods

Claim cards (`claims/*.yaml`) fix `property`, `scope`, `quantifier`, `success_criterion`, and `assumptions` (plus source_ref/version and original vs operational interpretation). Classical baselines live in `baselines/*.md` as authored notes, not Claim fields. **Pins** (`artifacts/pins.yaml`) freeze revisions and hashes. **Evidence** uses Finding labels; `scripts/check_evidence.py` fails closed if `evidence/index.yaml` disagrees with findings or supporting refs are missing. The checker's metric-binding scope is limited: reported metrics must agree with linked Run observations when keys overlap—it does not re-execute harnesses.

R1 (SoundGate). Rebuild and sweep the paper recovery-barrier harness at pinned source `6568403`. Compare policies `original`, `strict`, and `barrier` over the published histories. Success for reproduction: barrier `prohibited_effects == 0` with original showing the defect. Full `raw.json` is gitignored; committed `evidence/r1/raw.json.gz` supports local inspection. Recorded `raw_sha256` is integrity only.

R2 (EvidenceGraph). The harness **runs at `a5ea5c9`**, which publishes `incident_stress.py`. Paper baseline `efbd419` is the pre-extension tree (no stress script); baseline *behavior* is represented by replaying the authors' later-published stress pack at `a5ea5c9`. Label `reproduced` when all contract checks pass.

R3 (two distinct protocols). Legacy **r3-matched-decision-model** uses shared `cases.yaml` decision-model inputs; ClassicalGate emits allow/deny per case. BMO decisions require isomorphic observations—never package pass counts. When any required BMO case is unmatched, that comparison Finding is **`unresolved`**. Derived **r3-ledger-claim-v1** separately compares ClassicalLedgerGate (SQLite) to native `ClaimOrdinaryRetainedGrantFirstEditTx` under pin `4b717d7` with fail-closed native qualification (manifest, exit 0, adapter identity, git-archive pin inventory, bridge hashes, archived log).

R4 (three-way conclusions). Worlds A (grant active) and B (grant revoked) share an identical old receipt. Taxonomy in `src/swarm_witness/conclusion_classes.py`: **missing_evidence** → Finding `unresolved`; **failed_enforcement** → `contradicted_in_scope`; **supported** → `reproduced`. Blind/irrelevant cases must stay missing_evidence. A broken gate that allows under an observed revoked grant must be failed_enforcement. Correct allow/deny under **authentic** observed grant is supported. Misbound observations (available but false vs ground truth) are scored against truth and cannot yield unqualified `permission_justified`. Protocol: `experiments/evidence_sufficiency/`.

R5 (multi-principal transfer). Protocol `r5-swarm-audit-v1` uses an in-memory sequential ledger with principals A/B/C (`src/swarm_witness/swarm_audit.py`). Cases cover missing peer observation, sequential contended spend (correct and broken-decision-without-effect), sequential A→B then B→C assignee hops, and ledger-bound cross-agent receipt misuse. Each case uses the same three-way classes.

4. Results

Labels from `evidence/index.yaml` / findings only: R1 **reproduced**, R2 **reproduced**, R3 decision-model **unresolved**, R3 ledger-claim-v1 **classical_baseline_matches** (bounded L1), R4 **reproduced**, R5 **reproduced**. For R1/R2, `reproduced` means pinned independent replication. For R4/R5 it means synthetic protocol conformance under self-authored deterministic cases—not external replication. Metrics cite `evidence/*/summary.json`.

4.1 R1 — SoundGate recovery barrier (`reproduced`)

Fresh local rebuild+sweep; `all_checks_pass: true`; `raw_sha256=51b4acd6…`. Histories per policy: **153**.

Policy	cases	prohibited_effect	refused_before_ack	supported	fresh_work
original	153	**143**	4	6	149
strict	153	0	**147**	6	6
barrier	153	**0**	0	**153**	153

Barrier restores acknowledge-and-restrict across restart under the published histories; original loses restrictions (143 prohibited effects). This establishes reproducibility within tested conditions, not a new recovery principle. Classical recover-before-admit is an authored baseline interpretation (`baselines/r1_wal_recover_before_admit.md`).

4.2 R2 — Receipt ≠ attribution (`reproduced`)

Harness at `a5ea5c9`; `condition_by_seed_checks` **36**, passed **36**; seeds 0,1,2; `all_contract_checks_pass: true`; `source_tree_sha256=2962568f…`.

Metric (per seed)	seed 0	seed 1	seed 2
old_rule wrong attributions on relay	1	1	1
safer supported on relay	0	0	0
safer correct on relay	0	0	0
clean withheld without exclusivity	12	12	12
shared_tokens correct	0	0	0
false_exclusivity confident errors	1	1	1

Relay seed-0 outcomes: ambiguous 10, unmatched 2. Safer default withholds attribution without exclusivity; old rule errs once per seed on relay. Bounded reproduction of a published information-quality failure; classical no-implicit-exclusivity remains authored interpretation (`baselines/r2_no_implicit_exclusivity.md`).

4.3 R3 — Retained grant vs ClassicalGate (`unresolved`)

Matched **decision-model** comparison: **0/4** isomorphic BMO observations; `missing_bmo_cases: 4`. Comparison Finding `r3-f-comparison` label **`unresolved`** — comparison not performed / incomplete. ClassicalGate arm completed all four cases under the decision model (`state`, `request`, `grant_version`); BMO arm records `execution_status=missing` / `missing_isomorphic_observation` for each. Selected preferred Go tests on the pinned BMO revision remain available as package metadata and are **not** used as decisions or as a substitute comparative label.

Case	ClassicalGate	BMO decision	BMO status
retained_start_claim_relaunch	deny	—	missing
rejected_execute	deny	—	missing
active_grant_execute	allow	—	missing
stale_grant_version	deny	—	missing

4.4 R3 — Ledger claim L1 (`classical_baseline_matches`)

Derived protocol **r3-ledger-claim-v1** is distinct from §4.3. Across six synthetic cases (T01–T06), ClassicalLedgerGate and the native BMO L1 bridge at pin `4b717d7` recorded equal dispositions and property-relevant projections (grant_spent, counts, call_id sets) at every step, and both arms satisfied the declared one-use / reopen / rollback properties. Finding `r3-ledger-claim-v1-comparison` label **`classical_baseline_matches`** under fail-closed native qualification. Cross-arm equality does **not** claim native grant/epoch hash identity parity with synthetic classical aliases. This does not establish general BMO equivalence, multi-agent handoff, concurrency, or process-restart correctness. Selected run: `evidence/r3/ledger_claim_v1/2026-11-07-ledger-v1-repair2/`.

4.5 R4 — Three-way conclusions (`reproduced` = synthetic conformance)

Synthetic cases under `r4-evidence-sufficiency-v1` exercise all three classes, including **bounded handoff** and a **misbound negative control**: when the supplied grant observation disagrees with ground truth, the conclusion is `failed_enforcement` and `justified_permission_claim` is false (not `supported`). Blind Worlds A/B stay

missing_evidence; broken revoked-still-allows is failed_enforcement; authentic correct gates are supported. Cross-study map: baselines/three_way_conclusion_map.md. Finding r4-f-evidence-sufficiency label **reproduced**; all_checks_pass: true.

4.6 R5 — Multi-principal transfer (reproduced = synthetic conformance)

Ten cases under r5-swarm-audit-v1 with principals **A, B, C** in a sequential shared-grant model: missing peer observation (2); failed enforcement on broken post-spend allow and foreign-receipt allow under revoke, both without unauthorized effect commit (2); supported sequential contention, A→B then B→C assignee hops, and ledger-bound cross-agent receipt deny (6). Finding r5-f-swarm-audit label **reproduced**; agent_count: 3; model: sequential_shared_grant_multi_principal. This establishes taxonomy transfer to a small sequential multi-principal model—not concurrent swarm dynamics or live BMO mesh safety.

5. Discussion

Synthesis. Swarm Witness reproduces two established safety findings (R1/R2), compares a bounded BMO ledger property against a conventional reference monitor (R3-L1), and tests three-way evidence classification on synthetic authorization and multi-principal scenarios (R4/R5). That is the supported claim surface.

Implications. Multi-principal settings amplify evidence gaps and enforcement failures; an audit method must refuse peer activity, foreign receipts, or misbound observations as substitute warrant. Swarm Witness makes that refusal executable.

Limitations (summary). ClassicalGate is a decision-model null; legacy R3 is incomplete; ledger-claim equality is projection-scoped; R4/R5 are synthetic sequential protocols (not live concurrent agents); R5 does not establish emergent swarm dynamics. Full Limitations appendix follows.

Future work. Isomorphic decision-model R3 probes; grant/epoch alias maps; genuine contested concurrent handoffs; continuous baseline-vs-claim diffs.

Appendix A. Limitations and Dual-Use / Ethical Considerations

What each result does and does not establish

R1 (reproduced). Establishes that, on the pinned SoundGate recovery-barrier harness (source revision 6568403), the barrier policy yields prohibited_effects = 0 across 153 histories while the original policy yields 143 prohibited effects, with 147 strict refusal-startup refusals under the strict policy. Does **not** establish that every agent admission gate is restart-safe, that SoundGate is free of other defect classes, or that recover-before-admit is a new multi-agent invention (that classical reduction remains an authored baseline question under test). Committed evidence/r1/raw.json.gz supports local inspection of the sweep artifact (gzip -dc evidence/r1/raw.json.gz > evidence/r1/raw.json); recorded raw_sha256 is an integrity check and **does not replace** inspecting raw.json / analysis case records.

R2 (reproduced). Establishes that the stress pack executed at harness rev a5ea5c9 passes 36/36 condition-by-seed checks: old-rule relay wrong attributions are 1 per seed; safer relay supported attributions are 0; clean runs withhold all 12 attributions without exclusivity declarations. Paper baseline efbd419 is the pre-extension tree (no incident_stress.py); its baseline *behavior* is represented by replaying the authors' later-published stress pack at a5ea5c9, not by running a missing harness at efbd419. Does **not** establish that all receipt or ledger systems are safe under relay/copy, or that exclusivity declarations themselves cannot be gamed outside the modeled threat surface.

R3 decision-model (`unresolved`). Establishes that the matched ClassicalGate decision-model comparison against BMO is **incomplete**: 0/4 required cases have isomorphic BMO observations, so the comparison Finding is `unresolved` (comparison not performed). ClassicalGate is a **decision-model** null (`state`, `request`, `grant_version`)—not a restart-durability claim. Selected preferred Go tests on the pinned BMO revision are package metadata only and do not fill decisions or substitute for a matched comparative label. Does **not** establish ClassicalGate ↔ BMO API equivalence, assumption-dependent affinity, or a methodological reduction of BMO's durable-grant story to classical controls.

R3 ledger-claim-v1 (`classical_baseline_matches`, bounded). Establishes equal dispositions and property-relevant projections (grant_spent, counts, call_id sets) across T01–T06 for ClassicalLedgerGate vs native `ClaimOrdinaryRetainedGrantFirstEditTx` at pin `4b717d7` under fail-closed native qualification. Does **not** claim synthetic classical grant/epoch aliases equal native durable hashes, full event-trace observational equivalence, live execution safety, process-restart correctness, concurrency behavior, or general BMO equivalence. Legacy decision-model R3 remains separately unresolved.

R4 three-way conclusions (`reproduced` = synthetic conformance). Establishes that the audit method can classify synthetic cases into missing_evidence, failed_enforcement, and supported, including a misbound-observation negative control. Does **not** establish live BMO behavior or cryptographic receipt authenticity. The Finding label `reproduced` here means protocol conformance, not independent external replication.

R5 multi-principal transfer (`reproduced` = synthetic conformance). Establishes that the same three-way method classifies a sequential in-memory three-principal model covering shared-grant contention, A→B then B→C assignee hops, and ledger-bound receipt misuse, distinguishing wrong decisions from committed effects. Does **not** establish concurrent worker execution, live BMO mesh equivalence, cryptographic multiparty authorization, or unbounded population-scale swarm safety.

Finite experiments and checker scope

Pinned histories, seeds `{0,1,2}`, and preferred Go tests are finite. Passing them does not imply equivalence across versions, storage backends, concurrent writers outside the model, or production deployments. Labels apply to the stated scope on the claim cards only. `scripts/check_evidence.py` binds reported metrics to linked Run observations when keys overlap; it does not re-execute harnesses or prove metric completeness beyond that binding.

Dual-use / adversarial use of the audit method

An executable claim-audit method can help **defenders** find weak gates and attribution gaps—and can also help **adversaries** locate the same gaps (restart races, possession-as-attribution, retained-metadata relaunch). We publish **tests, pins, matched baselines, and labels**, not exploit kits, weaponized harnesses, or production credentials. Reproduction artifacts are scoped to already-public or author-local preexisting trees (`PREEXISTING.md`).

Preexisting code vs weekend analysis

Oct 25 scaffold paths and the external BMO tree at pinned `bmo_root` / `git_rev` predate this sprint and were not modified for R3 beyond read-only `go test`. Weekend work is the Swarm Witness schema, pins, claim cards, baselines, experiment wrappers, evidence records, index

checker, and this report. Judges should score the audit contribution, not treat preexisting BMO surfaces as sprint inventions.